

Lab -1

Input.c

```
#include <stdio.h>
```

```
int main() {  
    int NewZea_Land=10;  
    int Dhaka2, Dhaka_213123, BD24, France, Bangladesh, if2, _1, England;  
    printf("The number is %d", number);  
}
```

Lex1.l

```
/*Definition Section */
```

```
%{  
#include <stdio.h>  
int keywords=0;  
int identifiers=0;  
%}
```

```
/*Rules Section */
```

```
%%  
(int|printf) {  
    keywords++;  
    printf("%s is a keyword \n", yytext);  
}
```

```
[a-zA-Z_][a-zA-Z_0-9]* {  
    identifiers++;  
    printf("%s is a identifier \n", yytext);  
}
```

```
. {}
```

```
%%
```

```
/* User code */
```

```
int main() {  
    yyin=fopen("input.c", "r");  
    yylex();  
    printf("Total number of keywords: %d \n", keywords);  
    printf("Total number of identifiers: %d \n", identifiers);  
}
```

```

}

int yywrap() {
    return 1;
}

```

Lab - 2

Lex1.1

```

/*Definition Section */
%{
#include <stdio.h>

%}

/*Rules Section */
%%
x+ {
    printf("Matched one or more occurrences of x: %s \n", yytext);
}
x+y+ {
    printf("Matched one or more occurrences of x and y: %s\n", yytext);
}

xyz {
    printf("Matched xyz exactly: %s \n", yytext);
}

. {}

%%

/* User code */

int main() {
    yyin=fopen("input.c", "r");
    yylex();
}

int yywrap() {

```

```

    return 1;
}

```

Lab -3

```

/* Definition Section */
%{
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include "bison.tab.h"
}%
/* Rules Section */
%%
[0-9]*\.[0-9]+ { yylval.frac=atof(yytext); return FRACTION; }
"+" { return PLUS; }
"-" { return MINUS; }
"*" { return MULTIPLY;}
"/" { return DIVISION;}
"(" {return LEFTPAREN;}
")" {return RIGHTPAREN;}
\n { return EOL; }
[ \t\r]+ {}
. { printf("Unknown character: %s\n", yytext);}
%%
int yywrap(void) { return 1; }

```

Bison part

```

/* Definition Section */
%{
#include <stdio.h>
#include <string.h>
int yylex(void);
int yyerror(const char *s);
extern FILE *yyin;
}%
%union {
float frac;
}
/* Token */

```

```

%token EOL
%token <frac> FRACTION
%token PLUS
%type <frac> exp2
%token MINUS
%token MULTIPLY
%left PLUS MINUS
%left MULTIPLY
%left DIVISION
%token LEFTPAREN
%token RIGHTPAREN
%type <frac> fact2
%%
input:

```

```

|input line
;
line:
    exp2 EOL {printf ("%f \n", $1);}
| exp2 {printf ("%f \n", $1);}
| EOL {}
;

```

```

exp2:
    exp2 PLUS exp2 { $$ = $1 + $3; }
| exp2 MINUS exp2 { $$ = $1 - $3; }
| exp2 MULTIPLY exp2 { $$ = $1 * $3; }
| exp2 DIVISION exp2 { $$ = $1 / $3; }
| fact2
;

```

```

fact2:
    FRACTION { $$ = $1; }
| LEFTPAREN exp2 RIGHTPAREN { $$ = $2; }
;
%%
/* User Section */
int main(void) {
    yyin = fopen("lab1.txt", "r");
    int res=yyparse();
    fclose(yyin);
    return res;
}

```

```
int yyerror(const char *s) {  
    printf("ERROR: %s \n", s);  
    return 0;  
}
```

Lab-4

Code

```
#include <stdio.h>  
#include <string.h>  
  
int main() {  
    int t;  
    if (scanf("%d", &t) != 1) return 0;  
    while (t--) {  
        char s[105];  
        scanf("%s", s);  
        int n = strlen(s);  
        int has_pair = 0;  
        for (int i = 0; i + 1 < n; ++i) {  
            if (s[i] == s[i+1]) { has_pair = 1; break; }  
        }  
        if (has_pair) printf("1\n");  
        else printf("%d\n", n);  
    }  
    return 0;  
}
```